上海市东方医院
同济大学附属东方医院

DF Stem Cell

上海市东方医院生物样本库
Shanghai East Hospital Biobank

GEMPLE 捷易 BIOTECHNOLOGY

Whole Exome Sequencing Report (SNV)

Donor ID	ZB11AN	
Sample ID/Type	ZB11AN-UCBMC / SP1909243206	Somatic cell
	ZB11AN-B / SP1909243222	iPS cell
	ZB11AN-H / SP1909243233	iPS cell
Test content	Whole-exome sequencing was performed on paired somatic cells and iPS cells. By analyzing variants in known disease-causing genes, the pathogenic risk level of these samples for known Mendelian disorders and hereditary tumors was assessed. If mutations with high pathogenic risk levels were identified, the detailed information was listed in the appendix part.	
detection result	No pathogenic or suspected pathogenic variants consistent with the genetic pattern were identified, indicating a low risk of the donor having a known Mendelian genetic disorder or hereditary tumor.	
Test conclusion	According to the test results, the iPS cell samples were both determined to be qualified.	

Detector: Owen Zhi 郑思育

Assessor： 杨通 Yang Tong

上海捷易医学检验实验室 报告专用章

Report date： 2019-11-01

Appendix

I. List of pathogenic/likely pathogenic variants detected in this test

gene	Chromosome location (hg19)	Variant nomenclature	Function change	Hetero-zygosity	variant classification	Variation in the sample	Related diseases (#MIM, inheritance pattern)	Disease risk
AGK	chr7: 141313964	NM_018238.4:exon7: c.409C>T:p.R137X	nonsense mutation	heter	pathogenic	owned by all three samples	1) Sengers syndrome (212350,AR); 2) Type 38 cataract (614691, AR)	Carrier; low risk of disease

Note: 1) Mutation classification follows the ACMG guidelines;
2) AD-autosomal dominant inheritance, AR-autosomal recessive inheritance.

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CNV analysis report

Donor ID	ZB11AN	
Sample ID/Type	ZB11AN-UCBMC / SP1909243206	Somatic cell
	ZB11AN-B / SP1909243222	iPS cell
	ZB11AN-H / SP1909243233	iPS cell
Test content	Whole exome sequencing was performed on paired somatic cells and iPS cells. Copy number variations (CNVs) were identified using two bioinformatics algorithms, CNVkit and PanelCn. Variants were filtered and analyzed by leveraging authoritative public knowledge bases such as ClinVar (Clinical Variant Database), DGV (Disease Genome Variation), and Decipher (Disease Genome Variation). Pathogenicity assessment of CNVs was conducted according to the chromosomal variation classification guidelines. Samples without identified known or suspected pathogenic CNVs were deemed qualified, while those with clearly identified pathogenic or likely pathogenic CNVs received corresponding risk alerts.	
detection result	No known pathogenic or likely pathogenic copy number variations were detected.	
Test conclusion	According to the test results, the iPS cell sample was determined to be qualified.	

Detector: Owen Zhi　　Assessor ： Yang Tong　　Report date ： 2019-11-15

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